WORKED SOLUTION

STEP II 2022, Question 10

Mechanics •

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Part (i) is the standard trajectory equation, and the trick is to read it as a *quadratic in* $\tan\alpha$, using $\sec^2\alpha = 1 + \tan^2\alpha$. A target point can be hit exactly when that quadratic has a real root, so the whole question reduces to a discriminant condition. Written in the horizontal coordinates (x, y), that condition turns out to describe a disc, and every later part is a matter of reading off where a line meets a circle.

Solution

Throughout we model the projectile as a particle moving freely under gravity, with no air resistance, and take $0 < \theta < \frac{\pi}{2}$.

Part (i)

Take the horizontal distance and the height measured from the point of projection. At time t,

$$s = ut\cos\alpha, \qquad h = ut\sin\alpha - \tfrac{1}{2}gt^2.$$

The horizontal motion gives $t = \dfrac{s}{u\cos\alpha}$, and substituting,

$$h = s\tan\alpha - \frac{gs^2}{2u^2\cos^2\alpha},$$

as required.

Part (ii)

Let $P = (x, y, y\tan\theta)$ and write $s = \sqrt{x^2 + y^2}$ for its horizontal distance from the cannon. Since the cannon can point in any direction, we aim it in the vertical plane through P; the motion then takes place in that plane, and by part (i) we need an angle

α (possibly negative, since the cannon can point downwards) with

$$y\tan\theta = s\tan\alpha - \frac{gs^2}{2u^2}(1+\tan^2\alpha).$$

If $s = 0$ then P is the origin, which is reached by firing straight up, and the inequality below holds there because $\sec^2\theta > \tan^2\theta$. For $s > 0$, put $T = \tan\alpha$, which can be any real number. The condition is that

$$\frac{gs^2}{2u^2}T^2 - sT + \left(y\tan\theta + \frac{gs^2}{2u^2}\right) = 0$$

has a real root T, that is, its discriminant is non-negative:

$$s^2 \geq 4\cdot\frac{gs^2}{2u^2}\left(y\tan\theta + \frac{gs^2}{2u^2}\right) \iff u^4 \geq 2gu^2 y\tan\theta + g^2s^2,$$

after dividing by $s^2 > 0$ and multiplying by u^4. Dividing by g^2 and using $s^2 = x^2 + y^2$,

$$x^2 + y^2 + \frac{2u^2\tan\theta}{g}y \leq \frac{u^4}{g^2}.$$

Completing the square in y, and using $1+\tan^2\theta = \sec^2\theta$,

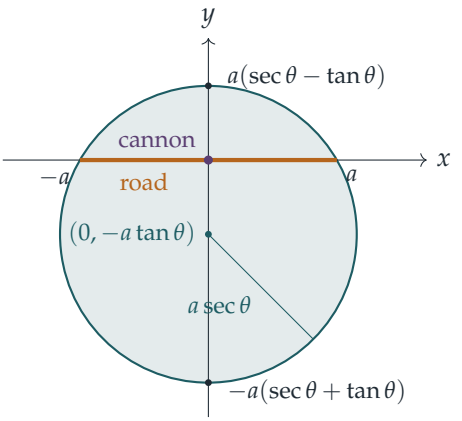
$$x^2 + \left(y + \frac{u^2\tan\theta}{g}\right)^2 \leq \frac{u^4}{g^2}(1+\tan^2\theta) = \frac{u^4\sec^2\theta}{g^2},$$

which is the required condition.

? **Ask yourself** Could the projectile hit the sloping ground *before* it reaches P?

No. In the vertical plane of the shot, the height of the projectile minus the height of the ground is a quadratic in the horizontal distance with negative leading coefficient (the parabola minus a straight line), and it vanishes at the cannon and at P. So it is positive in between: the projectile stays above the ground until it lands at P.

From now on write $a = u^2/g$. The reachable points are those whose horizontal projection lies in the disc with centre $(0, -a\tan\theta)$ and radius $a\sec\theta$.



The figure (drawn with $\theta = 30°$) shows the reachable region seen from above; "up the plane" is the positive y direction.

Part (iii)

Directly up the plane means $x = 0$, $y > 0$. The condition becomes $(y + a\tan\theta)^2 \leq a^2\sec^2\theta$, and since $y + a\tan\theta > 0$ this is

$$y \leq a(\sec\theta - \tan\theta).$$

A point on the plane with horizontal coordinate y lies at distance $y\sec\theta$ from the origin, measured along the plane. So the furthest point up the plane is at distance

$$a\sec\theta(\sec\theta - \tan\theta) = a\,\frac{1-\sin\theta}{\cos^2\theta} = a\,\frac{1-\sin\theta}{(1-\sin\theta)(1+\sin\theta)} = \frac{u^2}{g(1+\sin\theta)},$$

as required.

Directly down the plane means $x = 0$, $y < 0$. Now the condition gives $y \geq -a(\sec\theta + \tan\theta)$, so the furthest point down the plane is at distance

$$a\sec\theta(\sec\theta + \tan\theta) = a\,\frac{1+\sin\theta}{\cos^2\theta} = \frac{u^2}{g(1-\sin\theta)}$$

from the cannon.

Part (iv)

The road is the line $y = 0$ (which lies in the plane, at height 0). Putting $y = 0$ in the condition gives

$$x^2 \leq a^2\sec^2\theta - a^2\tan^2\theta = a^2,$$

so the reachable road is $-a \leq x \leq a$, a length of $\dfrac{2u^2}{g}$, the same as on level ground.

Now place the cannon at $(0, c, c\tan\theta)$. Its distance from the road (the x-axis) is $\sqrt{c^2 + c^2\tan^2\theta} = |c|\sec\theta$, so $r = |c|\sec\theta$. Everything in part (ii) depends only on positions *relative to the cannon*, and the plane looks the same from every point on it, so we can use the condition with y replaced by $y - c$. The road ($y = 0$) is reached where

$$x^2 \leq a^2\sec^2\theta - (a\tan\theta - c)^2.$$

? **Ask yourself** Should the cannon be placed above the road or below it?

The reachable length is $2\sqrt{a^2\sec^2\theta - (a\tan\theta - c)^2}$, which is largest when the subtracted square is zero, that is when $c = a\tan\theta > 0$. So the cannon goes *up* the plane from the road, exactly far enough that the road passes through the centre of the reachable disc, and it then cuts a diameter. The best distance is

$$r = a\tan\theta\sec\theta = \frac{u^2\sin\theta}{g\cos^2\theta},$$

and the maximum length of road is the diameter of the disc,

$$2a\sec\theta = \frac{2u^2}{g\cos\theta}.$$

Checks

- With $\theta = 0$ (level ground) part (ii) becomes $x^2 + y^2 \leq (u^2/g)^2$, the familiar maximum range u^2/g in every direction; both answers in (iii) reduce to u^2/g, and in (iv) $r = 0$ with length $2u^2/g$.
- The up-slope answer $u^2/\big(g(1+\sin\theta)\big)$ is the standard maximum range up an inclined plane, a good sign that the disc is right.
- With $u = 12\ \mathrm{m\,s^{-1}}$, $g = 9.8\ \mathrm{m\,s^{-2}}$ and $\theta = 0.5$ rad, a computer scan over all firing angles found the maximum range up the slope to be 9.932 m and down the slope 28.226 m, matching $u^2/\big(g(1\pm\sin\theta)\big)$; the best road length with the moved cannon was 33.49 m $= 2u^2/(g\cos\theta)$.

Common slips

- Treating (x, y) as distances along the plane. They are horizontal coordinates, so a distance up the plane is $y\sec\theta$, not y.
- Forgetting that the cannon may point below the horizontal: $\tan\alpha$ ranges over all real numbers, which is why the discriminant condition is the whole story.
- Placing the moved cannon below the road. The disc's centre is down the slope from the cannon, so the cannon must sit above the road.

Beyond STEP

The disc in part (ii) is a slice of the *envelope* of all possible trajectories, the "paraboloid of safety" $z = \dfrac{u^2}{2g} - \dfrac{g(x^2+y^2)}{2u^2}$; intersecting this paraboloid with any plane gives a conic, here a circle in projection. Envelopes of families of curves are a natural first-year topic in both mechanics and differential equations. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see projectile problems treated with this kind of geometric care.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.